Chapter 3: Composition of Substances and Solutions

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Lecture 3

Part I

Formula Mass and the Mole Concept

Outline of Part I: Formula Mass and the Mole Concept

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Why Count Atoms?

Chemical reactions occur between atoms and molecules. To control a reaction, we need to know **how many** of each type of atom or molecule we are working with.

We cannot count atoms one by one — a grain of sand contains $\sim 10^{18}$ atoms.

Instead, we count atoms **by mass** using the concept of the **mole**.

- A baker measures flour by **mass** rather than counting individual grains.
- A chemist measures by **mass** rather than counting individual particles.

Formula Mass

Definition: Formula Mass

The **formula mass** of a substance is the sum of the average atomic masses of all atoms represented in its chemical formula, expressed in atomic mass units (amu).

For covalent molecules, the formula mass is called the **molecular mass**.
For ionic compounds, it is simply called the **formula mass** (since no discrete molecules exist).

Atomic masses are found on the periodic table.
Sum **all** atoms — including those inside parentheses multiplied by the subscript outside.

Calculating Formula Mass: Molecular Compounds

Formula Mass of Chloroform, $CHCl_3$ — (chem2e_ch3_ex3.1)

$$M(\mathrm{CHCl_3}) = 1(12.011) + 1(1.008) + 3(35.450)$$
$$= 12.011 + 1.008 + 106.350$$

C: $1 \times 12.011 = 12.011$
H: $1 \times 1.008 = 1.008$
Cl: $3 \times 35.450 = 106.350$

$$\boxed{M(\mathrm{CHCl_3}) = 119.37 \text{ amu}}$$

Formula Mass of Ibuprofen, $C_{13}H_{18}O_2$ — (chem2e_ch3_ex3.2)

$$M = 13(12.011) + 18(1.008) + 2(15.999) = 156.143 + 18.144 + 31.998 = \boxed{206.28 \text{ amu}}$$

Calculating Formula Mass: Ionic Compounds

Formula Mass of Aluminum Sulfate, $Al_2(SO_4)_3$ — (chem2e_ch3_ex3.3)

Expand the formula: $Al_2S_3O_{12}$ (2 Al, 3 S, 12 O)

$$M = 2(26.982) + 3(32.065) + 12(15.999)$$

$$= 53.964 + 96.195 + 191.988$$

$$\boxed{M = 342.15 \text{ amu}}$$

The parentheses mean the (SO_4) group appears **3 times**: 3 S and $3 \times 4 = 12$ O atoms total.

Check Your Understanding

Without calculating, which compound has the larger formula mass?

(a) CO or CO_2

(b) $NaCl$ or $MgCl_2$

(c) H_2O or H_2O_2

Come to lecture to see the answer.

Outline of Part I: Formula Mass and the Mole Concept

The Mole: A Counting Unit

We use counting units to group large numbers of things.

Familiar Counting Units

a **dozen** $= 12$ of something

a **gross** $= 144$ of something

a **mole** $= 6.022 \times 10^{23}$ of something

Definition: Mole

One **mole** (mol) is the amount of substance containing exactly $6.02214076 \times 10^{23}$ elementary entities (atoms, molecules, ions, formula units, etc.).

This number is **Avogadro's number**: $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$.

Why Is Avogadro's Number So Large?

N_A is defined so that one mole of an element has a mass (in grams) numerically equal to its atomic mass (in amu).

Thing counted	**1 particle (amu)**	**1 mole (g)**
Sulfur atom	32.066 amu	32.066 g
Carbon atom	12.011 amu	12.011 g
Water molecule	18.015 amu	18.015 g

$1 \text{ amu} \equiv 1 \text{ g/mol}$ — the definition is designed to make this exact correspondence.

Molar Mass

Definition: Molar Mass (M)

The **molar mass** of a substance is the mass in grams of exactly one mole of that substance.

Numerically equal to the formula mass in amu, but with units of $\frac{\text{g}}{\text{mol}}$.

Molar Mass of Water

1 mol H_2O contains 2 mol H and 1 mol O:

$$\begin{aligned} M_{\mathrm{H_2O}} &= 2\,M_{\mathrm{H}} + 1\,M_{\mathrm{O}} \\ &= 2\left(1.008\ \tfrac{\text{g}}{\text{mol}}\right) + 1\left(15.999\ \tfrac{\text{g}}{\text{mol}}\right) = \boxed{18.015\ \tfrac{\text{g}}{\text{mol}}} \end{aligned}$$

Molar mass of Na_2SO_4: $M = 2(22.990) + 32.065 + 4(15.999) = \boxed{142.04\ \frac{\text{g}}{\text{mol}}}$

Check Your Understanding

True or False — explain your reasoning:

"One mole of iron (Fe) and one mole of gold (Au) contain the same number of atoms, but do not have the same mass."

Come to lecture to see the answer.

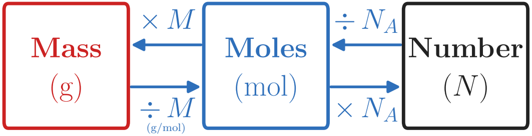
Outline of Part I: Formula Mass and the Mole Concept

Formula Mass

The Mole and Molar Mass

Mole-Mass-Number Conversions

The Conversion Pathway



- mass $\xrightarrow{\div M}$ mol   mol $\xrightarrow{\times M}$ mass
- mol $\xrightarrow{\times N_A}$ atoms   atoms $\xrightarrow{\div N_A}$ mol

Key formulas:

$$n = \frac{m}{M} \quad m = n\,M \quad N = n\,N_A$$

Example: Mass to Moles — (chem2e_ch3_ex3.4)

How Many Moles of Potassium Are in $4.7\,\mathrm{g}$ of K?

$M_\mathrm{K} = 39.10\ \frac{\mathrm{g}}{\mathrm{mol}}$

$$n = \frac{m}{M} = 4.7\,\mathrm{g}\left(\frac{1\ \mathrm{mol\ K}}{39.10\ \mathrm{g}}\right)$$

$$\boxed{n = 0.12\ \mathrm{mol\ K}}$$

Units: $\dfrac{\mathrm{g}}{\mathrm{g/mol}} = \mathrm{mol}$ ✓

Answer has 2 sig. figs. because $m = 4.7$ g has 2 sig. figs.

Example: Moles to Mass — (chem2e_ch3_ex3.5)

What Is the Mass of 9.2×10^{-4} mol of Argon?

$M_{\mathrm{Ar}} = 39.948\ \frac{\mathrm{g}}{\mathrm{mol}}$

$$m = n \times M$$

$$m = (9.2 \times 10^{-4}\ \mathrm{mol})(39.948\ \tfrac{\mathrm{g}}{\mathrm{mol}})$$

$$\boxed{m = 0.037\ \mathrm{g}}$$

Units: $\mathrm{mol} \times \mathrm{g}/\mathrm{mol} = \mathrm{g}$ ✓

Example: Mass to Atoms — (chem2e_ch3_ex3.6)

How Many Cu Atoms Are in $5.00\,\mathrm{g}$ of Copper?

$M_{\mathrm{Cu}} = 63.546\ \frac{\mathrm{g}}{\mathrm{mol}}$

$$N = 5.00\,\mathrm{g}\left(\frac{1\ \mathrm{mol\ Cu}}{63.546\ \mathrm{g}}\right)\left(\frac{6.022\times10^{23}\ \mathrm{atoms}}{1\ \mathrm{mol}}\right)$$

$$\boxed{N = 4.74 \times 10^{22}\ \mathrm{Cu\ atoms}}$$

Two conversion factors chained together:

(1) g → mol
(2) mol → atoms

Example: Moles from a Chemical Formula

Find the Number of Moles of Ag in 2.1 Moles of AgCl

AgCl has 1 Ag and 1 Cl per formula unit.
$\therefore$ there is 1 mol Ag per 1 mol AgCl.

$$n_{\mathrm{Ag}} = 2.1 \text{ mol AgCl} \left(\frac{1 \text{ mol Ag}}{1 \text{ mol AgCl}} \right)$$

$$\boxed{n_{\mathrm{Ag}} = 2.1 \text{ mol Ag}}$$

The **subscript** in a formula gives the mole ratio of each element to one formula unit.

Example: Counting Atoms via Avogadro's Number

Find the Number of Oxygen Atoms in 0.92 mol Na_2SO_4

Na_2SO_4 contains 4 O atoms per formula unit, so 4 mol O per mol Na_2SO_4.

$$N_{\mathrm{O}} = 0.92 \text{ mol } \mathrm{Na_2SO_4} \left(\frac{4 \text{ mol O}}{1 \text{ mol } \mathrm{Na_2SO_4}}\right) \left(\frac{6.022{\times}10^{23} \text{ atoms}}{1 \text{ mol O}}\right)$$

$$\boxed{N_{\mathrm{O}} = 2.2 \times 10^{24} \text{ O atoms}}$$

Two conversion factors: mole ratio from the formula, then Avogadro's number.

Check Your Understanding

A sample of water, H_2O, contains 0.500 mol.

Without calculating, which quantity is larger?

(a) The number of H atoms, or the number of O atoms?

(b) The number of O atoms, or the number of H_2O molecules?

Come to lecture to see the answer.

Part II

Percent Composition and Chemical Formulas

Outline of Part II: Percent Composition and Chemical Formulas

Percent Composition

Empirical and Molecular Formulas

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Percent Composition

Empirical and Molecular Formulas

Percent Composition by Mass

Definition: Percent Composition

The **percent composition** of a compound is the mass percentage of each element in the compound:

$$\%\ X = \frac{\text{mass of } X \text{ in 1 mol compound}}{M_{\text{compound}}} \times 100\%$$

Example: Percent H and O in Water

$$\%\ \mathrm{H} = \frac{2(1.008\ \mathrm{g/mol})}{18.015\ \mathrm{g/mol}} \times 100\% = \boxed{11.19\%}$$

$$\%\ \mathrm{O} = \frac{15.999\ \mathrm{g/mol}}{18.015\ \mathrm{g/mol}} \times 100\% = \boxed{88.81\%}$$

All percentages must sum to 100%.
$11.19 + 88.81 = 100.00\%$

Example: Percent Composition from Experimental Data — (chem2e_ch3_ex3.9)

A 12.04 g Sample Contains 7.34 g C, 1.85 g H, and 2.85 g N. Find the Percent Composition.

$$\%\ \mathrm{C} = \frac{7.34\ \mathrm{g}}{12.04\ \mathrm{g}} \times 100\% = \boxed{61.0\%}$$

$$\%\ \mathrm{H} = \frac{1.85\ \mathrm{g}}{12.04\ \mathrm{g}} \times 100\% = \boxed{15.4\%}$$

$$\%\ \mathrm{N} = \frac{2.85\ \mathrm{g}}{12.04\ \mathrm{g}} \times 100\% = \boxed{23.7\%}$$

Check: $61.0 + 15.4 + 23.7 = 100.1\%$

The 0.1% discrepancy is due to rounding. This is acceptable.

Example: Mass Percent from Formula

Find the Mass Percent of Sulfur in Na_2SO_4

$$M_{\mathrm{Na_2SO_4}} = 2(22.990) + 32.065 + 4(15.999) = 142.04\ \frac{\mathrm{g}}{\mathrm{mol}}$$

$$\%\ \mathrm{S} = \frac{1(32.065\ \mathrm{g/mol})}{142.04\ \mathrm{g/mol}} \times 100\%$$

$$\boxed{\%\ \mathrm{S} = 22.6\%}$$

In 0.92 mol $\mathrm{Na_2SO_4}$:
mass of S
$= 0.92\ \mathrm{mol} \times 32.065\ \mathrm{g/mol}$
$= 29.5\ \mathrm{g}$
total mass
$= 0.92 \times 142.04 = 130.7\ \mathrm{g}$
$\%\mathrm{S} = 29.5/130.7 \times 100\% = 22\%$
$(2\ \mathrm{s.f.})\ \checkmark$

Check Your Understanding

Aspirin has the molecular formula $C_9H_8O_4$.

Which element has the **highest** mass percent, without calculating?

Hint: Consider how many atoms of each element there are and their atomic masses.

Come to lecture to see the answer.

Outline of Part II: Percent Composition and Chemical Formulas

Percent Composition

Empirical and Molecular Formulas

Empirical vs. Molecular Formula

Empirical Formula

Gives the **simplest whole-number ratio** of atoms in a compound.

Examples:

- CH_2O (ratio C:H:O = 1:2:1)
- Fe_2O_3
- HO

Molecular Formula

Gives the **actual** number of each type of atom in one molecule.

Examples:

- $C_6H_{12}O_6$ (glucose; empirical: CH_2O)
- Fe_2O_3 (same as empirical)
- H_2O_2 (empirical: HO)

Molecular formula = (empirical formula) $\times$ n, where n is a positive integer.

Determining Empirical Formula from Data

Procedure

1. **Assume 100 g** of compound (percent → grams directly).
2. **Convert grams to moles** for each element: $n_i = m_i/M_i$.
3. **Divide all moles** by the smallest value to get ratios.
4. **Round to whole numbers**. If a ratio ≈ 1.5, multiply all by 2; if ≈ 1.33, multiply by 3.

Finding molecular formula: if the molar mass M is known,

$$n = \frac{M_{\text{compound}}}{M_{\text{empirical formula}}}$$

Then multiply all subscripts in the empirical formula by n.

Example: Empirical Formula from Masses — (chem2e_ch3_ex3.11)

A Hematite Sample Contains 34.97 g Fe and 15.03 g O.
Find the Empirical Formula.

$$n_{\mathrm{Fe}} = \frac{34.97\ \mathrm{g}}{55.845\ \frac{\mathrm{g}}{\mathrm{mol}}} = 0.6261\ \mathrm{mol}$$

$$n_{\mathrm{O}} = \frac{15.03\ \mathrm{g}}{15.999\ \frac{\mathrm{g}}{\mathrm{mol}}} = 0.9394\ \mathrm{mol}$$

Divide by smallest (0.6261):

$$\mathrm{Fe} : \frac{0.6261}{0.6261} = 1.000 \quad \mathrm{O} : \frac{0.9394}{0.6261} = 1.500$$

$\times 2$: Fe_2O_3

Empirical formula: $\boxed{Fe_2O_3}$

A ratio of ≈ 1.5 means the true ratio is $3:2$ after multiplying by 2.

Example: Empirical Formula from Masses

A Sample Contains 5.08 g Zn, 1.60 g P, and 3.32 g O.
Find the Empirical Formula.

$$n_{\mathrm{Zn}} = \frac{5.08\ \mathrm{g}}{65.38\ \frac{\mathrm{g}}{\mathrm{mol}}} = 0.0777\ \mathrm{mol}$$

$$n_{\mathrm{P}} = \frac{1.60\ \mathrm{g}}{30.974\ \frac{\mathrm{g}}{\mathrm{mol}}} = 0.0517\ \mathrm{mol}$$

$$n_{\mathrm{O}} = \frac{3.32\ \mathrm{g}}{15.999\ \frac{\mathrm{g}}{\mathrm{mol}}} = 0.208\ \mathrm{mol}$$

Divide by 0.0517:

Zn: 1.50, P: 1.00, O: 4.02 $\xrightarrow{\times 2}$ $Zn_3P_2O_8$

$Zn_3P_2O_8 = \boxed{Zn_3(PO_4)_2}$

(zinc phosphate)

Zinc always has charge +2; phosphate is PO_4^{3-}.

Zn: $1.50 \approx \frac{3}{2}$, so multiply all by 2.

Example: Molecular Formula from Empirical — (chem2e_ch3_ex3.13)

Nicotine: 74.02% C, 8.710% H, 17.27% N; Molar Mass 162.3 g/mol.
Find the Molecular Formula.

Assume 100 g:

$$n_C = \frac{74.02 \text{ g}}{12.011} = 6.163 \text{ mol}$$
$$n_H = \frac{8.710 \text{ g}}{1.008} = 8.641 \text{ mol}$$
$$n_N = \frac{17.27 \text{ g}}{14.007} = 1.233 \text{ mol}$$

Divide by 1.233: C: 4.99, H: 7.01, N: 1.00
Empirical: C_5H_7N ($M_{\text{emp}} = 81.13$)

$$n = \frac{162.3}{81.13} \approx 2.00$$

Molecular formula: $\boxed{C_{10}H_{14}N_2}$
(Nicotine — found in tobacco plants)

Check Your Understanding

A compound contains only C, H, and O. Its empirical formula is CH_2O and its molar mass is 180.2 g/mol.

What is its molecular formula?

(You may need to calculate the empirical formula mass.)

Come to lecture to see the answer.

Part III

Chemical Equations and Stoichiometry

Outline of Part III: Chemical Equations and Stoichiometry

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Chemical Equations

Definition: Chemical Equation

A **chemical equation** represents a chemical reaction using chemical formulas and symbols.

Reactants appear on the **left**; products appear on the **right**; an arrow ($\rightarrow$) separates them.

State Symbols

- (s) = solid (l) = liquid (g) = gas (aq) = dissolved in water

Law of Conservation of Mass

Atoms are neither created nor destroyed in a chemical reaction.

$\therefore$ the number of each type of atom must be **equal** on both sides of the equation.

An equation satisfying this condition is said to be **balanced**.

How to Balance a Chemical Equation

Procedure

1. Write the **unbalanced** equation with correct formulas.
2. Count each type of atom on both sides.
3. Place **coefficients** (whole numbers) in front of formulas to balance.
4. Never change subscripts — that would change the identity of the substance.
5. Verify: equal atoms of every element on both sides.

The coefficients represent the **mole ratios** of reactants and products in the reaction.

Example: Balancing — Electrolysis of Water

Balance: $H_2O \rightarrow H_2 + O_2$

1. Unbalanced: $H_2O \rightarrow H_2 + O_2$
2. Count O: 1 left, 2 right $\Rightarrow$ imbalanced.
3. Place 2 in front of H_2O:
   $2\,H_2O \rightarrow H_2 + O_2$
4. Count H: 4 left, 2 right $\Rightarrow$ place 2 before H_2:
   $2\,H_2O \rightarrow 2\,H_2 + O_2$
5. H: $4 = 4$ ✓; O: $2 = 2$ ✓

$$2\,H_2O(l) \rightarrow 2\,H_2(g) + O_2(g)$$

This is the **electrolysis** of water: electrical energy splits water into hydrogen and oxygen gas.

Example: Balancing — Combustion of Methane

Balance: $CH_4 + O_2 \rightarrow CO_2 + H_2O$

1. C: $1 = 1$ ✓
2. H: 4 left, 2 right $\Rightarrow$ place 2 before H_2O: $CH_4 + O_2 \rightarrow CO_2 + 2\,H_2O$
3. O: 2 left, $2 + 2 = 4$ right $\Rightarrow$ place 2 before O_2: $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$
4. Check: C 1=1 ✓; H 4=4 ✓; O 4=4 ✓

$$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$$

This balanced equation is the **basis** for all stoichiometric calculations on the following slides.

Check Your Understanding

Which of these is a correctly balanced equation?

(A) $N_2 + H_2 \rightarrow NH_3$

(B) $N_2 + 3\,H_2 \rightarrow 2\,NH_3$

(C) $2\,N_2 + 3\,H_2 \rightarrow 2\,NH_3$

Come to lecture to see the answer.

Outline of Part III: Chemical Equations and Stoichiometry

Using Mole Ratios as Conversion Factors

The **coefficients** in a balanced equation give the mole-to-mole ratios between any two species in the reaction.

For $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$:

1 mol CH_4 reacts with 2 mol O_2

1 mol CH_4 produces 1 mol CO_2

1 mol CH_4 produces 2 mol H_2O

📝 **General stoichiometry path:**

$$\underbrace{m_A}_{\text{given}} \xrightarrow{\div M_A} n_A \xrightarrow{\text{mole ratio}} n_B \xrightarrow{\times M_B} \underbrace{m_B}_{\text{desired}}$$

Example: Mass–Mass Calculation

I Want $7.0\,\mathrm{g}$ $H_2O(g)$. What Mass of $CH_4(g)$ Is Needed? (O_2 in Excess)

Balanced: $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$

$$m_{CH_4} = 7.0\,\mathrm{g}\;H_2O \left(\frac{1\ \mathrm{mol}\ H_2O}{18.015\ \mathrm{g}}\right)\left(\frac{1\ \mathrm{mol}\ CH_4}{2\ \mathrm{mol}\ H_2O}\right)\left(\frac{16.043\ \mathrm{g}}{1\ \mathrm{mol}\ CH_4}\right)$$

$$\boxed{m_{CH_4} = 3.1\ \mathrm{g}}$$

Three conversion factors:

(1) g $H_2O \rightarrow$ mol H_2O

(2) mol $H_2O \rightarrow$ mol CH_4

(3) mol $CH_4 \rightarrow$ g CH_4

Example: Mass of Excess Reactant Remaining

50 g of O_2 Is Present Initially. How Much Remains After Producing 7.0 g H_2O?

Balanced: $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$

$$m_{O_2\text{ used}} = 7.0\text{ g H}_2\text{O}\left(\frac{1\text{ mol H}_2\text{O}}{18.015\text{ g}}\right)\left(\frac{2\text{ mol O}_2}{2\text{ mol H}_2\text{O}}\right)\left(\frac{31.998\text{ g}}{1\text{ mol O}_2}\right)$$

$$m_{O_2\text{ used}} \approx 12\text{ g}$$

$$m_{O_2\text{ remaining}} = 50\text{ g} - 12\text{ g}$$

$$\boxed{m_{O_2\text{ remaining}} = 38\text{ g}}$$

Find mass **consumed** from stoichiometry, then subtract from the initial amount.

Check Your Understanding

For the reaction: $2\,H_2 + O_2 \rightarrow 2\,H_2O$

If 4.00 mol H_2 reacts completely, how many moles of H_2O are produced?

Can you answer this without calculating? Explain using mole ratios.

Come to lecture to see the answer.

Outline of Part III: Chemical Equations and Stoichiometry

Theoretical Yield, Actual Yield, Percent Yield

Definitions

- **Theoretical yield**: maximum mass of product predicted by stoichiometry, assuming 100% conversion of limiting reactant.
- **Actual yield**: mass of product actually collected from the experiment.
- **Percent yield**: ratio of actual to theoretical, expressed as a percentage.

$$\%\ \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Yield is always expressed as a **mass** (or sometimes moles).
Percent yield can **never exceed 100%**.

Why Is Actual Yield < Theoretical Yield?

Common Reasons

- Reaction does not go to completion (equilibrium is reached before all reactant is consumed).
- **Side reactions** consume some of the reactant to form undesired products.
- Product is **lost during transfer or purification**.
- Measurement or experimental errors.

In synthetic chemistry, a percent yield above 90% is considered excellent.
Many industrial reactions operate at 50–80%.

Example: Percent Yield

I Burn 3.1 g CH_4 and Collect 6.2 g H_2O (Theoretical Yield: 7.0 g). Find the Percent Yield.

Balanced: $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$

$$\%\ \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

$$\%\ \text{yield} = \frac{6.2\ \text{g}}{7.0\ \text{g}} \times 100\%$$

$$\boxed{\%\ \text{yield} = 88\%}$$

Theoretical yield (7.0 g) was computed from 3.1 g CH_4 by stoichiometry.

Actual yield (6.2 g) was measured experimentally.

Check Your Understanding

In an experiment, a student obtains 4.85 g of a product. The theoretical yield is 6.02 g.

(a) What is the percent yield?

(b) A classmate claims a percent yield of 105%. Is this possible? What would it mean?

Come to lecture to see the answer.

Part IV

Solution Concentrations

Outline of Part IV: Solution Concentrations

Molarity

Other Units for Solution Concentrations

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Molarity

Other Units for Solution Concentrations

Solutions: Key Terms

Definition: Solution

A **solution** is a **homogeneous mixture** — its composition and properties are uniform throughout its entire volume.

- **Solvent**: the component present in the *greater* amount; the "dissolving medium."
- **Solute**: the component present at a *lower* concentration; the "dissolved" substance.

In saltwater:

- **Solvent** = water (most abundant)
- **Solute** = $NaCl$ (less abundant)

An **aqueous solution** is one in which the solvent is water.

Concentration

Concentration is the quantitative measure of the relative amounts of solute and solvent.

Molarity

Definition: Molarity (M)

Molarity is the number of moles of solute dissolved per liter of solution:

$$M = \frac{n_{\text{solute}}}{V_{\text{solution}}} \qquad \left[\frac{\text{mol}}{\text{L}} = \text{M}\right]$$

Example: Sucrose in a Soft Drink — (chem2e_ch3_ex3.15)

A 355-mL can contains 0.133 mol sucrose.

$$M = \frac{0.133 \text{ mol}}{0.355 \text{ L}} = \boxed{0.375 \text{ M}}$$

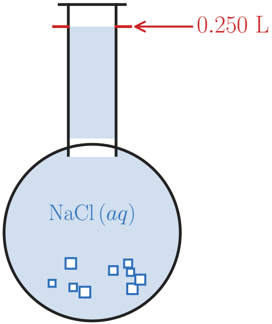
Molarity uses volume of **solution**, not volume of **solvent**. Adding solute changes the volume.

Example: Grams from Molarity — (chem2e_ch3_ex3.16)

How Many Grams of NaCl Are in 0.250 L of a 5.30 M NaCl Solution?

$M_{\mathrm{NaCl}} = 58.44\ \frac{\mathrm{g}}{\mathrm{mol}}$

$$n_{\mathrm{NaCl}} = M \times V$$
$$= 5.30\ \frac{\mathrm{mol}}{\mathrm{L}} \times 0.250\ \mathrm{L}$$
$$= 1.325\ \mathrm{mol}$$
$$m_{\mathrm{NaCl}} = 1.325\ \mathrm{mol} \times 58.44\ \frac{\mathrm{g}}{\mathrm{mol}}$$
$$\boxed{m_{\mathrm{NaCl}} = 77.4\ \mathrm{g}}$$



$n = M \times V$
$m = n \times M_{\mathrm{molar}}$
Chain: M, V $\rightarrow$ mol $\rightarrow$ g

Dilution

Dilution is the addition of solvent to lower the concentration of a solution.

The moles of solute do **not change**:

$$n_1 = n_2 \implies C_1 V_1 = C_2 V_2$$

Subscript 1 = before dilution; subscript 2 = after dilution.

Example — (chem2e_ch3_ex3.17)

0.850 L of 5.00 M $Cu(NO_3)_2$ is diluted to 1.80 L. What is the new concentration?

$$C_2 = \frac{C_1 V_1}{V_2} = \frac{(5.00\ \mathrm{M})(0.850\ \mathrm{L})}{1.80\ \mathrm{L}} = \boxed{2.36\ \mathrm{M}}$$

Check Your Understanding

A student dilutes 11 mL of 0.45 M HBr to a final volume of 41 mL.

(a) Does the number of moles of HBr change?

(b) What is the new concentration?

Come to lecture to see the answer.

Outline of Part IV: Solution Concentrations

Mass Percentage

Definition: Mass Percentage

$$\%\text{ mass} = \frac{m_{\text{solute}}}{m_{\text{solution}}} \times 100\%$$

Also written % mass, % (w/w), percent by mass.

Example: Glucose in Spinal Fluid — (chem2e_ch3_ex3.22)

A 5.0 g spinal fluid sample contains 3.75 mg glucose.

$$\%\text{ glucose} = \frac{0.00375\text{ g}}{5.0\text{ g}} \times 100\% = \boxed{0.075\%}$$

Mass percentage is **temperature-independent** (uses masses, not volumes).

Volume Percentage and Mass-Volume Percentage

Volume Percentage (% vol)

$$\% \text{ vol} = \frac{V_{\text{solute}}}{V_{\text{solution}}} \times 100\%$$

Used when both solute and solvent are liquids.

Example: 70% isopropanol (70 mL isopropanol per 100 mL solution)

Mass-Volume Percentage (% m/v)

$$\% \text{ m/v} = \frac{m_{\text{solute}}(\text{g})}{V_{\text{solution}}(\text{mL})} \times 100\%$$

Example: Physiological saline = 0.9% (m/v)
= 0.9 g NaCl per 100 mL solution

Volume percentages are common in consumer products (rubbing alcohol, wine, vinegar) and clinical settings (IV solutions).

Parts per Million and Parts per Billion

Definitions

- **Parts per million (ppm)**: $\text{ppm} = \frac{m_{\text{solute}}}{m_{\text{solution}}} \times 10^6$
- **Parts per billion (ppb)**: $\text{ppb} = \frac{m_{\text{solute}}}{m_{\text{solution}}} \times 10^9$

Example: Lead in Drinking Water — (chem2e_ch3_ex3.25)

EPA limit for lead in drinking water: 15 ppb.
In 300 mL ($\approx$ 300 g) of water at 15 ppb:

$$m_{\text{Pb}} = 15 \times 10^{-9} \times 300\ \text{g} = 4.5 \times 10^{-6}\ \text{g} = \boxed{4.5\ \mu\text{g}}$$

For **aqueous** solutions: $1\ \text{ppm} \approx 1\ \frac{\text{mg}}{\text{L}}$ (since $\rho_{\text{water}} \approx 1\ \text{g/mL}$).

Concentration Units: Summary

Unit	Symbol	Formula	Units
Molarity	M	$n_{\mathrm{sol}}/V_{\mathrm{soln}}$	mol/L
Mass %	%	$(m_{\mathrm{sol}}/m_{\mathrm{soln}}) \times 100$	%
Volume %	%vol	$(V_{\mathrm{sol}}/V_{\mathrm{soln}}) \times 100$	%
Mass-vol %	%m/v	$(m_{\mathrm{sol}}/V_{\mathrm{soln}}) \times 100$	g/100 mL
ppm	ppm	$(m_{\mathrm{sol}}/m_{\mathrm{soln}}) \times 10^6$	dimensionless
ppb	ppb	$(m_{\mathrm{sol}}/m_{\mathrm{soln}}) \times 10^9$	dimensionless

Check Your Understanding

A solution is labeled "0.9% NaCl (m/v)."

(a) What does this mean in terms of grams per volume?

(b) Would you expect this concentration to change if the solution is heated? Why or why not?

Come to lecture to see the answer.

Summary: Chapter 3

- **Formula mass**: sum of atomic masses of all atoms in a formula (amu).
- **Molar mass**: formula mass in g/mol; bridges atomic and macroscopic scales via Avogadro's number ($N_A = 6.022 \times 10^{23}$ mol^{-1}).
- **Percent composition**: $\%\,\mathrm{X} = (m_X/M_{\mathrm{compound}}) \times 100\%$.
- **Empirical formula**: simplest whole-number atom ratio (from mass data or % composition).
- **Molecular formula**: actual atoms per molecule; found from empirical formula and molar mass.
- **Balanced equations**: obey conservation of mass; coefficients give mole ratios.
- **Stoichiometry**: mass A $\rightarrow$ mol A $\rightarrow$ mol B $\rightarrow$ mass B using mole ratios.
- **Percent yield**: (actual/theoretical) $\times$ 100%; always $\leq$ 100%.
- **Molarity** (M): mol solute / L solution; dilution: $C_1 V_1 = C_2 V_2$.
- Other units: mass%, volume%, mass-volume%, ppm, ppb.